

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Review/Challenge Questions

1. A series has a sum if what sequence has a limit?

In order for a series to have a sum it must be the case that $\lim_{n \rightarrow \infty} s_n = L$. So, the sequence of partial sums s_n must have a limit.

2. Consider an infinite series $\sum_{n=1}^{\infty} a_n$.

- a) Suppose that $\sum_{n=1}^{\infty} a_n = 3$. What can you say about the sequence $\{a_n\}$?

Since the series converges it must be the case that $\lim_{n \rightarrow \infty} a_n = 0$.

- b) Suppose that $\lim_{n \rightarrow \infty} a_n = \frac{1}{3}$, what can you say about $\sum_{n=1}^{\infty} a_n$?

Every convergent series must have $\lim_{n \rightarrow \infty} a_n = 0$, so this series must diverge.

3. Find the sum of the following series or explain why the sum does not exist.

- a) $1 + \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots + \frac{1}{3^n} + \dots$

We can rewrite this series as $\sum_{n=1}^{\infty} \frac{1}{3^n} = \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n = \sum_{n=1}^{\infty} \frac{1}{3} \left(\frac{1}{3}\right)^{n-1}$. This is a geometric series with $r = \frac{1}{3}$ and

since $|r| < 1$ this series converges. Specifically, $\sum_{n=1}^{\infty} \frac{1}{3} \left(\frac{1}{3}\right)^{n-1} = \frac{1/3}{1-(1/3)} = \frac{1/3}{2/3} = \frac{1}{2}$

b) $1+2+4+8+\dots+2^n+\dots$

We can rewrite this series as $\sum_{n=1}^{\infty} 2^n = \sum_{n=1}^{\infty} 2(2)^{n-1}$. This is a geometric series with $r = 2$ and since $|r| \geq 1$ this series diverges.

c) $\sum_{i=4}^{\infty} \frac{3}{i}$

This series can be rewritten as $3 \sum_{i=4}^{\infty} \frac{1}{i}$. We see that this resulting series is just the Harmonic series with a few terms missing and so we can conclude that this series diverges just like the Harmonic series.

4. Determine the sum of the following series.

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{9} + \dots + \left(\frac{1}{2^n} + \frac{1}{3^n} \right) + \dots$$

The series can be rewritten as $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{9} + \dots + \left(\frac{1}{2^n} + \frac{1}{3^n} \right) + \dots = \sum_{n=1}^{\infty} \left(\frac{1}{2^n} + \frac{1}{3^n} \right)$.

We can rewrite this series as two separate series because each series will be a geometric series which converges.

Note that

$$\begin{aligned} \sum_{n=1}^{\infty} \left(\frac{1}{2^n} + \frac{1}{3^n} \right) &= \sum_{n=1}^{\infty} \left(\frac{1}{2^n} \right) + \sum_{n=1}^{\infty} \left(\frac{1}{3^n} \right) = \sum_{n=1}^{\infty} \left(\frac{1}{2} \right)^n + \sum_{n=1}^{\infty} \left(\frac{1}{3} \right)^n = \sum_{n=1}^{\infty} \frac{1}{2} \left(\frac{1}{2} \right)^{n-1} + \sum_{n=1}^{\infty} \frac{1}{3} \left(\frac{1}{3} \right)^{n-1} \\ &= \frac{1/2}{1-(1/2)} + \frac{1/3}{1-(1/3)} = 1 + \frac{1/3}{2/3} = 1 + \frac{1}{2} \\ &= \frac{3}{2} \end{aligned}$$

5. Can you use the Ratio Test to determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^4}$. Why or why not?

No. Note that $\lim_{n \rightarrow \infty} \frac{\left(\frac{1}{(n+1)^4}\right)}{\left(\frac{1}{n^4}\right)} = \lim_{n \rightarrow \infty} \frac{n^4}{(n+1)^4} = 1$, so the Ratio Test is inconclusive here.

6. ($n!$ versus 2^n) Show that $\sum_{n=1}^{\infty} \frac{2^n}{n!}$ converges. What does this tell you about $\lim_{n \rightarrow \infty} \frac{2^n}{n!}$? Which sequence goes faster to zero, $\left\{\frac{1}{2^n}\right\}$ or $\left\{\frac{1}{n!}\right\}$?

We note that the series contains only non-zero terms and

$$\lim_{n \rightarrow \infty} \frac{\left(\frac{2^{n+1}}{(n+1)!}\right)}{\left(\frac{2^n}{n!}\right)} = \lim_{n \rightarrow \infty} \frac{2^{n+1} n!}{2^n (n+1)!} = \lim_{n \rightarrow \infty} \frac{2}{n+1} = 0 < 1$$

Therefore, by the Ratio Test, this series must converge.

Since the series converges it must be the case that $\lim_{n \rightarrow \infty} \frac{2^n}{n!} = 0$ and this implies that the sequence $a_n = \frac{1}{n!}$ must approach 0 faster than $b_n = \frac{1}{2^n}$.

7. Determine whether the following series are (absolutely or conditionally) convergent, or divergent.

a) $\sum_{n=1}^{\infty} \frac{n^2 + 1}{n^2 + 4}$

We notice here that $\lim_{n \rightarrow \infty} \frac{n^2 + 1}{n^2 + 4} = 1$ and therefore by the Test for Divergence this series must diverge.

b) $\sum_{n=2}^{\infty} \frac{1}{n^2 - 1}$

Consider the series $\sum_{n=2}^{\infty} \frac{1}{n^2}$. Note that both series contain only positive terms and

$$\lim_{n \rightarrow \infty} \frac{\left(\frac{1}{n^2 - 1} \right)}{\left(\frac{1}{n^2} \right)} = \lim_{n \rightarrow \infty} \frac{n^2}{n^2 - 1} = 1.$$

Since the result of this limit is positive and finite then, by the LCT, both series must have the same behavior. Since $\sum_{n=2}^{\infty} \frac{1}{n^2}$ is a convergent p-series then $\sum_{n=2}^{\infty} \frac{1}{n^2 - 1}$ must also converge. Note that since

$$\sum_{n=2}^{\infty} \left| \frac{1}{n^2 - 1} \right| = \sum_{n=2}^{\infty} \frac{1}{n^2 - 1} \text{ and so the series converges absolutely.}$$

c) $\sum_{n=1}^{\infty} \frac{n!}{n^2}$

We note that the series contains only non-zero terms and

$$\lim_{n \rightarrow \infty} \frac{\left(\frac{(n+1)!}{(n+1)^2} \right)}{\left(\frac{n!}{n^2} \right)} = \lim_{n \rightarrow \infty} \frac{n^2 (n+1)!}{(n+1)^2 n!} = \lim_{n \rightarrow \infty} \frac{n^2 (n+1)}{(n+1)^2} = \lim_{n \rightarrow \infty} \frac{n^2}{(n+1)} = \infty > 1$$

Therefore, by the Ratio Test, this series must diverge.

d) $\sum_{n=1}^{\infty} \frac{(n!)^n}{(n^4)^n}$

$$\text{We note that } \lim_{n \rightarrow \infty} \sqrt[n]{\frac{(n!)^n}{(n^4)^n}} = \lim_{n \rightarrow \infty} \frac{n!}{n^4} = \infty > 1.$$

Therefore, by the Root Test, this series must diverge.

e) $\sum_{n=1}^{\infty} (-1)^n \frac{\ln(n)}{n}$

This is an alternating series with $a_n = \frac{\ln(n)}{n}$. Note that if $f(x) = \frac{\ln(x)}{x}$ then $f'(x) = \frac{1 - \ln(x)}{x^2}$ which is decreasing for $x \geq e$. Therefore, a_n is decreasing for $n \geq 3$. We also see that $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0$ and so $\lim_{n \rightarrow \infty} a_n = 0$. Therefore, by the AST, the series $\sum_{n=3}^{\infty} (-1)^n \frac{\ln(n)}{n}$ converges and so $\sum_{n=1}^{\infty} (-1)^n \frac{\ln(n)}{n}$ must also converge since it only adds 2 extra terms.

Note that $\sum_{n=1}^{\infty} \left| (-1)^n \frac{\ln(n)}{n} \right| = \sum_{n=1}^{\infty} \frac{\ln(n)}{n}$. If $f(x) = \frac{\ln(x)}{x}$, then we know that $f(x)$ is positive, continuous, and decreasing on $[3, \infty)$ and since

$$\int_3^{\infty} \frac{\ln(x)}{x} dx = \lim_{t \rightarrow \infty} \left[\int_3^t \frac{\ln(x)}{x} dx \right] = \lim_{t \rightarrow \infty} \left[\int_{\ln(3)}^{\ln(t)} u du \right] = \lim_{t \rightarrow \infty} \left[\frac{1}{2} \ln(t)^2 - \frac{1}{2} \ln(3)^2 \right] = \infty$$

we know by the Integral

Test that the series $\sum_{n=3}^{\infty} (-1)^n \frac{\ln(n)}{n}$ is absolutely divergent and therefore so is $\sum_{n=1}^{\infty} (-1)^n \frac{\ln(n)}{n}$.

Thus, the series $\sum_{n=1}^{\infty} (-1)^n \frac{\ln(n)}{n}$ is conditionally convergent.

f) $\sum_{n=1}^{\infty} \frac{3^n}{3^n - 2^n}$

Note that for all $n \geq 1$, $\frac{3^n}{3^n - 2^n} \geq \frac{3^n}{2^n} = \left(\frac{3}{2}\right)^n$. We see that the series $\sum_{n=1}^{\infty} \frac{3^n}{3^n - 2^n}$ and $\sum_{n=1}^{\infty} \left(\frac{3}{2}\right)^n$ are both series with non-negative terms and $\sum_{n=1}^{\infty} \left(\frac{3}{2}\right)^n$ is a divergent geometric series with $r = \frac{3}{2}$. Therefore, by the Comparison Test, the series $\sum_{n=1}^{\infty} \frac{3^n}{3^n - 2^n}$ must diverge.

8. For what values of x does the series absolutely converge?

a) $\sum_{n=0}^{\infty} (x-2)^n$

If $x = 2$ then the series simplifies to $\sum_{n=0}^{\infty} 0 = 0$ and is therefore convergent. It is also easy to see that this series would be absolutely convergent.

If $x \neq 2$ then the series will always contain non-zero terms and the from the Ratio Test we see that

$$\lim_{n \rightarrow \infty} \frac{|x-2|^{n+1}}{|x-2|^n} = \lim_{n \rightarrow \infty} |x-2| = |x-2|$$

If the result of this limit is less than 1 then the series converges absolutely.

Therefore we just need to determine for what values of x it is the case that $|x-2| < 1$.

We calculate that $|x-2| < 1 \rightarrow -1 < x-2 < 1 \rightarrow 1 < x < 3$. Therefore, if $1 < x < 3$ the series will converge absolutely.

b) $\sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!}$

If $x = 0$ then the series simplifies to $\sum_{n=0}^{\infty} 0 = 0$ and is therefore convergent. It is also easy to see that this series would be absolutely convergent.

If $x \neq 0$ then the series will always contain non-zero terms and the from the Ratio Test we see that

$$\lim_{n \rightarrow \infty} \frac{\left| \frac{x^{2(n+1)+1}}{(2(n+1)+1)!} \right|}{\left| \frac{x^{2n+1}}{(2n+1)!} \right|} = \lim_{n \rightarrow \infty} \frac{|x|^{2n+3} (2n+1)!}{(2n+3)! |x|^{2n+1}} = \lim_{n \rightarrow \infty} \frac{|x|^2}{(2n+3)(2n+2)} = |x|^2 \lim_{n \rightarrow \infty} \frac{1}{(2n+3)(2n+2)} = |x|^2 \cdot 0 = 0 < 1$$

The result of this limit is *always* less than 1 and so the series converges absolutely for all values of x .

9. Show that the following series diverges except for exactly one value of c and compute the sum of the series for that value of c .

$$\sum_{n=2}^{\infty} \frac{cn+1}{n^2-n}$$

Suppose that the series converges.

We note that
$$\sum_{n=2}^{\infty} \frac{cn+1}{n^2-n} = \sum_{n=2}^{\infty} \left(\frac{cn}{n^2-n} + \frac{1}{n^2-n} \right) = \sum_{n=2}^{\infty} \left(\frac{c}{n-1} + \frac{1}{n^2-n} \right).$$

We then split this series and investigate each resulting series separately.

We see that
$$\sum_{n=2}^{\infty} \left(\frac{c}{n-1} + \frac{1}{n^2-n} \right) = \sum_{n=2}^{\infty} \frac{c}{n-1} + \sum_{n=2}^{\infty} \frac{1}{n^2-n} = \sum_{n=1}^{\infty} \frac{c}{n} + \sum_{n=2}^{\infty} \frac{1}{n^2-n}.$$

Since $\sum_{n=1}^{\infty} \frac{c}{n}$ is the Harmonic series it diverges and the other series $\sum_{n=2}^{\infty} \frac{1}{n^2-n} = \sum_{n=2}^{\infty} \left(\frac{1}{n-1} - \frac{1}{n} \right)$ is a

telescoping series with $s_k = \left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{k-1} - \frac{1}{k} \right) = 1 - \frac{1}{k}$. Since $\lim_{k \rightarrow \infty} s_k = 1$ the

series $\sum_{n=2}^{\infty} \frac{1}{n^2-n}$ converges to 1.

Now, we note that if $\sum_{n=2}^{\infty} \frac{cn+1}{n^2-n}$ converges and $\sum_{n=2}^{\infty} \frac{1}{n^2-n}$ converges, then it must be the case that

$$\sum_{n=2}^{\infty} \left(\frac{cn+1}{n^2-n} - \frac{1}{n^2-n} \right) \text{ must converge. But, this implies that } \sum_{n=2}^{\infty} \left(\frac{cn+1}{n^2-n} - \frac{1}{n^2-n} \right) = \sum_{n=1}^{\infty} \frac{c}{n} \text{ must converge.}$$

The only way that this can happen is if $c = 0$.

Since the series of the 2nd fraction would converge (details below) then the series of $\frac{c}{n-1}$ would also have to converge else the given series could not converge. This implies that $c=0$ (see below)

Therefore, the given series only converges (specifically to 1) for $c = 0$, and diverges for all other values of c .